# Yijin Zeng

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## Summary

- **Machine Learning Scientist & PhD in Statistics and Machine Learning.**
- **Strong experience and knowledge in Deep Learning & Generative AI & Transformer.**
- **Visit my Personal Website for more information about me and my work.**

## Experience

**Expedia Machine Learning Scientist (London)** Oct.2025 – Current

- Work within the Revenue Optimisation & Data Science team to build, enhance, and maintain end-to-end machine learning pipelines for handling large-scale data and high-frequency requests.
- Explore and apply state-of-the-art research methods to continuously improve existing machine learning models. Enhance model evaluation frameworks by closing the gap between offline metrics and online performance. Monitor model health by automatically tracking key performance indicators and guardrail metrics.
- Translate complex business logic and constraints into actionable mathematical, statistical, or machine learning solutions.

**TESCO Technology Data Scientist PhD Internship (London)** Aug.2024 – Oct.2024

- Collaborated with forecasting and supply chain teams to improve forecasting accuracy for weekly product demand.
- Built an end-to-end data pipeline for model training, including automated feature selection, missing data imputation, outlier handling, feature scaling, and sequence generation using sliding windows.
- Designed and experimented with multiple deep neural network architectures, including RNNs and Transformers, for supply chain forecasting. Matched or exceeded Tesco's existing statistical approach, while requiring fewer workflows. Achieved up to a 23% improvement for specific product-store combinations and a statistically significant 7% average gain for products with limited historical data.

**IQVIA Data Analyst Internship (Beijing)** Jul.2019 – Sep.2019

- Worked in the data analytics team to analyze medical surveys and experimental datasets, extracting key insights for pharmaceutical products across new product launches, pricing, physician behaviors, patient journeys, and brand tracking.
- Preprocessed data for downstream machine learning models and subsequent statistical analysis through feature engineering, missing data imputation, and outlier detection.
- Designed data visualizations to effectively communicate complex results to technical and non-technical stakeholders.

## Education

- **Imperial College London**, London, UK — Oct. 2021– Oct. 2025
  PhD in Statistics and Machine Learning (StatML CDT, Imperial & University of Oxford)
  - *Fully funded by Roth Scholarship and EPSRC CDT*
- **Imperial College London**, London, UK — Oct. 2020 – Sep. 2021
  MSc in Statistics
  - *Degree classification: Distinction*
- **Beijing Institute of Technology**, Beijing, China — Sep. 2016 – Jun. 2020
  BSc in Mathematics
  - *Top 10% of cohort; Thesis with Distinction*

## Publications

- **Zeng Y**, Adams NM, Bodenham DA. MMD Two-sample Testing in the Presence of Arbitrarily Missing Data. *Transactions on Machine Learning Research (TMLR). 2025 Nov 20.*
  - *This work extends the popular maximum mean discrepancy (MMD) two-sample testing methods to the cases where data might be missing or partially observed.*
- **Zeng Y**, Adams NM, Bodenham DA. Exact Bounds of Spearman's footrule in the Presence of Missing Data with Applications to Independence Testing. *Electronic Journal of Statistics 19 (2), 6103-6140.*
  - *This work derives mathematical tight bounds for Spearman's footrule (a robust alternative to Spearman's rank correlation) in the presence of missing data, proposes efficient algorithms for computing the bounds, and applies the bounds for independence testing with missing data.*
- **Zeng Y**, Adams NM, Bodenham DA. On two-sample testing for data with arbitrarily missing values. arXiv preprint *arXiv:2403.15327. 2024 Mar 22.*
  - *A two-sample hypothesis testing framework for partially observed univariate data to detect location shifts, with no assumption on the values of missing data.*
- **Zeng Y**, Adams NM, Bodenham DA. Scale two-sample testing with arbitrarily missing data. *arXiv preprint arXiv:2509.20332. 2025 Sep 24.*
  - *A two-sample scale testing method in the presence of missing data without missing data assumptions.*

## Open Source Packages

- **Zeng Y**, Hazem M. Package 'sequential-speculative-decoding'.
  - *Python package implementing standard and sequential speculative decoding for accelerating LLM inference.*
- **Zeng Y**, Bodenham D, and Adams N. Package 'wmwm'.
  - *R and Python packages for performing the Wilcoxon-Mann-Whitney two sample hypothesis testing in the presence of missing data.*
- **Zeng Y**. Package 'bosfr'.
  - *R package for computing tight bounds of Spearman's footrule in the presence of missing data.*
- **Zeng Y**. Package 'abwm'.
  - *R package for performing the Ansari-Bradley two sample hypothesis testing in the presence of missing data.*

## Invited Talk

- StatML CDT Internal Seminar — Jan 2025
  - *I presented on modern techniques for handling missing data in machine learning at the StatML Centre for Doctoral Training (CDT).*
- Imperial Internal Statistics Seminar — June 2025
  - *I presented my PhD research on rank-based hypothesis testing in the presence of missing data at Imperial College London.*
- Royal Statistical Society (RSS) Conference 2026
  - *I will be presenting as an invited speaker at RSS 2026 in the session 'Modern Approaches for Missing and Incomplete Data Analysis'.*

## Selected Teaching Experience as GTA

- **MSc Courses:** Probability for Statistics, Probability Generative Models, Data Science, Statistical Clinics
- **BSc Courses:** Probability and Statistics, Signal Processing and Machine Learning, Data Mining and Analysis, Sets and Numbers, Applied Statistics and Business

## Programming & Tools

- Python, R, TensorFlow, PyTorch, Pandas, NumPy, Scikit-learn, Keras, HPC, Linux, Git, Azure ML, Databricks, Tableau